

GRADUATION PROJECT

Design and Implementation of an Intelligent Interview Simulation System (IISS)

TEAM INFORMATION

Team IISS

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ABOUT THE PROJECT

IISS replicates realistic job interview scenarios using a conversational AI engine and a virtual AI avatar. It performs multimodal behavioral analysis — tracking facial expressions, voice tone, speech clarity, and eye movements — to evaluate candidate performance. An anti-cheating framework integrates identity verification, multi-face and mobile phone detection to ensure fairness. The system instantly delivers data-driven reports highlighting strengths, weaknesses, and improvement recommendations.

PROBLEM STATEMENT

Existing interview preparation platforms are often limited in realism, personalization, and evaluation depth. Many systems rely on static question banks and do not adequately simulate real interview conditions. Furthermore, most platforms fail to analyze important verbal and non-verbal communication aspects such as facial expressions, tone of voice, and eye contact. As a result, candidates receive limited feedback on their actual performance, and recruiters lack objective, data-driven tools for fairly evaluating candidates. There is therefore a clear need for an intelligent system that provides realistic interview simulation and comprehensive multimodal performance analysis to improve both candidate preparation and hiring decisions.

SOLUTION

An AI-driven platform that offers:

- ▶ Facial emotion & eye-tracking analysis
- ▶ Voice tone & speech-to-text processing
- ▶ AI avatar for realistic interviews
- ▶ Anti-cheating: face/phone detection & ID verification
- ▶ Structured performance feedback & scoring
- ▶ Practicing and recruiting modes
- ▶ Monitor interviews by interactive dashboard
- ▶ Custom interviews by specified role, experience and question source (AI, custom dataset)

CONCLUSION

IISS combines AI, behavioral analysis, and conversational interaction to deliver a comprehensive interview simulation solution. Through facial emotion recognition, eye tracking, and speech analysis, it provides personalized feedback and performance evaluation. Security mechanisms ensure interview integrity, while data-driven insights support better hiring decisions — demonstrating the practical applicability of AI in recruitment and training.



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